

EEAIC assignment3

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1 Task

The feedback resistance R_F in the inverter-based amplifier with capacitive feedback is replaced with MOSFET-based pseudo-resistor configurations. These structures are essential for achieving the extremely high incremental resistance required to set the low-frequency cutoff of the amplifier without the massive area penalty of passive resistors.

Four possible configurations are analyzed as visible in *Figure 1*. The choice of using PMOS transistors is primarily a physical necessity due to the fact that each PMOS device lives in its isolated N-well and this gives us the freedom over the bulk terminal of each individual transistor without shorting the entire chip's substrate to a certain value.

Additionally, High-Threshold Voltage (HVT) devices were employed to minimize subthreshold conduction to mitigate the primary drawback of these configurations, which is leakage current.

A channel length of $L = 1\mu\text{m}$ was chosen during testing to provide high resistance and avoid Short Channel effects.

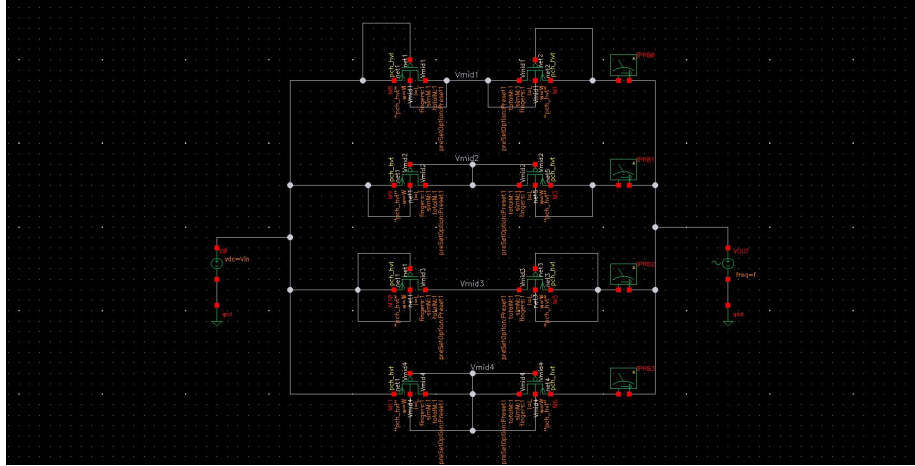


Figure 1: Testbench of the 4 possible configurations

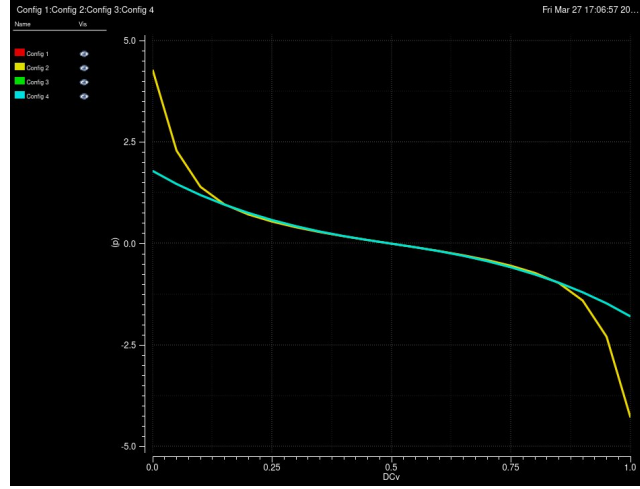


Figure 2: DC V-I characteristic

As observed in the DC I-V characteristic simulations from *Figure 2*, all possible configurations show a similar behavior when $|\Delta V| < 50mV$.

However, when comparing the four main configurations, a fundamental trade-off emerges between the stability of the DC bias and the saturation of the leakage current.

Advantages of Configurations 1 and 2

Configurations 1 and 2, characterized by a direct path through two series diodes, offer a behavior that more closely resembles an ideal resistor for small voltage swings. Their primary advantage lies in Stable DC Biasing within DC Servo Loops (DSL). In applications, potentially like ours, with wide output voltage swings, these configurations prevent the amplifier's input DC voltage from shifting because the parasitic diodes become forward-biased, drastically lowering the dynamic resistance. This provides an open path for the DSL to easily drive the necessary corrective current and maintain the DC bias. Unlike configurations 3 and 4, where current is bottlenecked to microscopic leakage levels causing the DSL to be unable to clear the accumulated charge at the input capacitor, consequently shifting the input DC voltage and eventually pushing the amplifier out of its proper operating region and failing the circuit's bias.

Furthermore, as shown in *Figure 3* and *Figure 4* they maintain a Symmetric Internal Node (V_{mid}), keeping the voltage balanced between V_{IN} and V_{OUT} , which ensures more predictable operation across the signal range.

Disadvantages of Configurations 1 and 2

The main drawback of these topologies is the Exponential Current Increase that occurs when the voltage difference across the device becomes large enough to forward-bias the parasitic diodes (*see Figure 2*). This leads to a Less Stable Resistance Current compared to other configurations. While Configurations 3 and 4 benefit from current saturation at the generation level of the space-charge re-

gion—providing a "bottleneck" that limits current, Configurations 1 and 2 suffer from uncontrolled exponential growth, which significantly lowers the equivalent resistance during large transients.

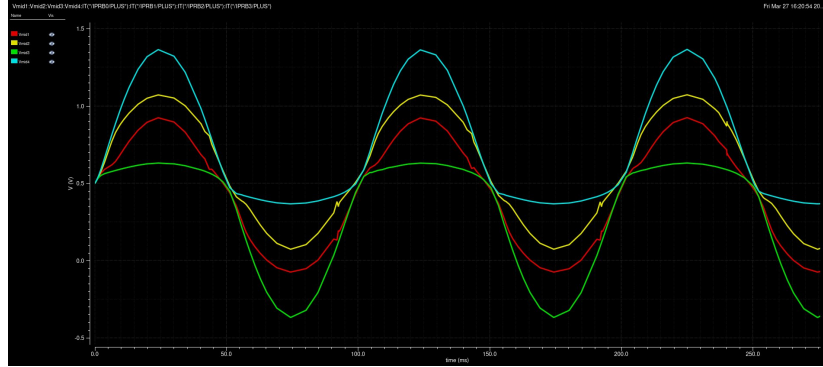


Figure 3: V_{mid} at each configuration with a very slow frequency $10Hz$

In conclusion, configurations 1 and 2 are preferred despite their less stable resistance current during large transients. By allowing forward-biased diode conduction, they provide an open path for charge neutralization. This ensures a fully functional DC Servo Loop (DSL) that maintains stable DC biasing and prevents input voltage drift, avoiding the severe saturation and biasing failures inherent in the current-limited topologies of configurations 3 and 4.

In particular, for our design Configuration 1 (*Figure 5*) was chosen.

Note that the current dimensions serve to demonstrate the expected qualitative behavior and the trade-offs between leakage tolerance and linear resistance; therefore, the transistors in Configuration 1 were not specifically sized to reach the target resistance of $10\text{ G}\Omega$.

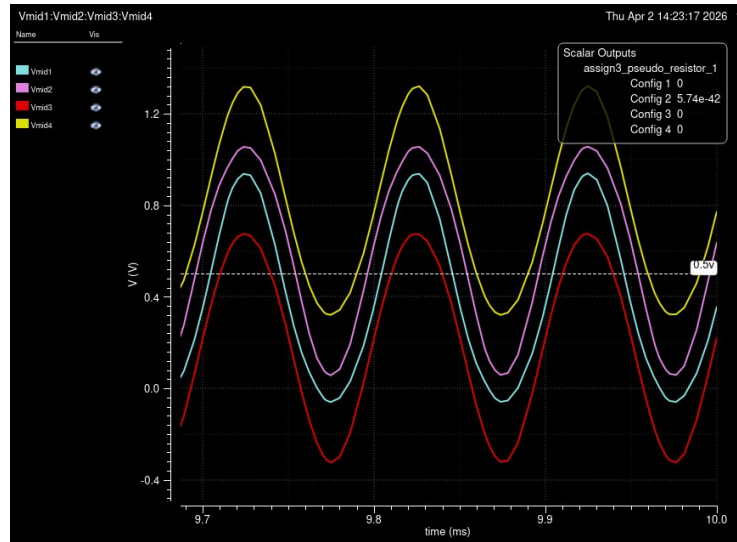


Figure 4: V_{mid} at each configuration with a high frequency $10kHz$

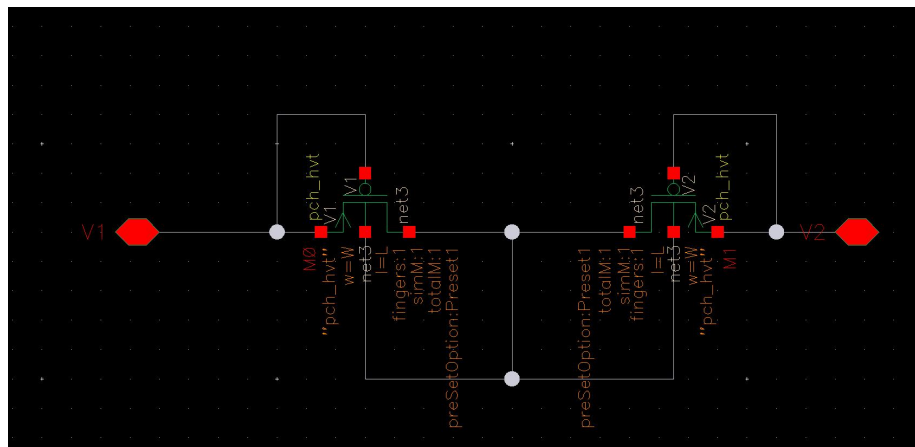


Figure 5: Configuration 1

2 Task

To eliminate low-frequency artifacts, specifically DC offset and $1/f$ noise, chopper stabilization is integrated into the amplifier design. This technique modulates the input signal to a higher frequency where low-frequency noise is negligible, amplifies it, and then demodulates it back to the baseband.

While a dedicated low-pass filter (LPF) would be required to properly remove all high-frequency harmonics, the limited bandwidth of our signal allows for effective recovery by integrating within its specific band. This stage is essential to filter the modulated flicker noise and other up-converted low-frequency artifacts, thereby ensuring accurate signal recovery at the output.

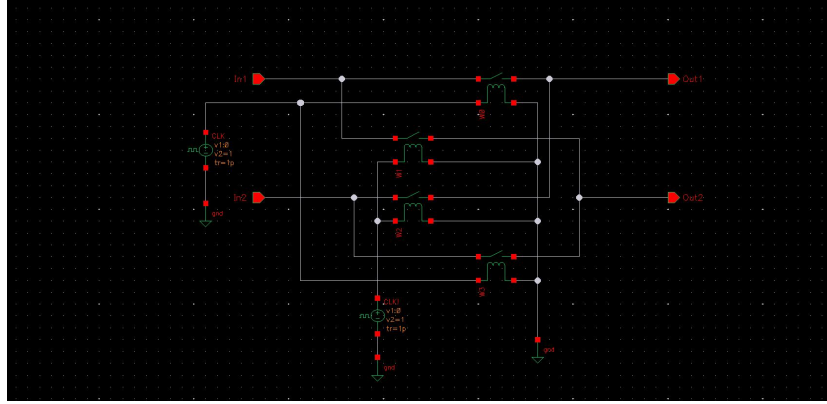


Figure 6: Schematic of the chopping block

The selection of the chopping frequency f_{ch} is critical to avoid both $1/f$ noise and spectral folding. To effectively shift the modulated signal entirely out of the flicker noise region, f_{ch} must exceed the amplifier's noise corner frequency (≈ 3 kHz, as shown in *Figure 8*) plus the maximum signal bandwidth ($f_{BW,max} = 5$ kHz).

Additionally, to prevent the modulated sidebands from overlapping with the baseband signal (avoiding aliasing-like interference), a Nyquist-like condition $f_{ch} > 2 \cdot f_{BW,max}$ must be satisfied. Although an initial design iteration utilized a chopping frequency of 15 kHz, to safely provide a wide margin for both criteria the final design was optimized to 20 kHz to ensure operation in a region dominated by thermal noise.

The fundamental drawback of this architecture is the requirement for high amplifier bandwidth ($BW|_{amp}$); indeed, to prevent the rounding of the modulated square waves and ensure the signal settles before each polarity flip, the amplifier bandwidth should ideally be at least 10 times the chopping frequency.

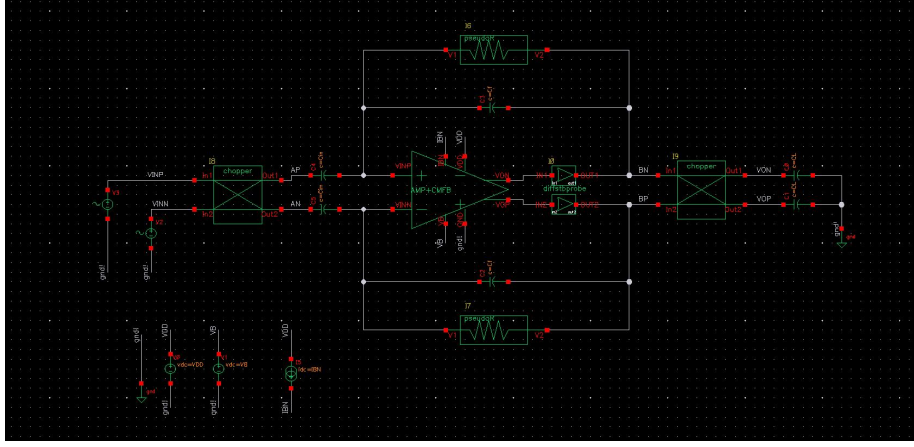


Figure 7: System with chopping blocks and pseudoresistors

While a higher f_{ch} improves noise performance, it simultaneously reduces the input impedance of the amplifier due to the periodic switching of the input capacitors ($R_{in} = \frac{1}{2f_{ch}C_{in}}$). The selected 20 kHz frequency represents an optimized trade-off between the noise floor and the intrinsic input impedance. However, it is important to emphasize that in high-precision designs, an impedance boosting loop is typically implemented.

The analysis sweeps the number of sidebands (N_{harm}) from 10 to 500 to ensure convergence. The results show that the noise value plateaus beyond a certain threshold (around 100 harmonics), which confirms that a sufficient number of harmonics are included to capture the full noise contribution of the periodic switching.

The transistors of Configuration 1 are sized to ensure that the transfer function remains consistent with the results obtained in *Exercise 2*. As shown in the Bode plots of *Figure 9*, the 20 dB gain is fully maintained. Specifically, the pseudo-resistor configuration successfully emulates the high-impedance feedback path, achieving the desired frequency response without the explicit implementation of a physical 10 G Ω resistor.

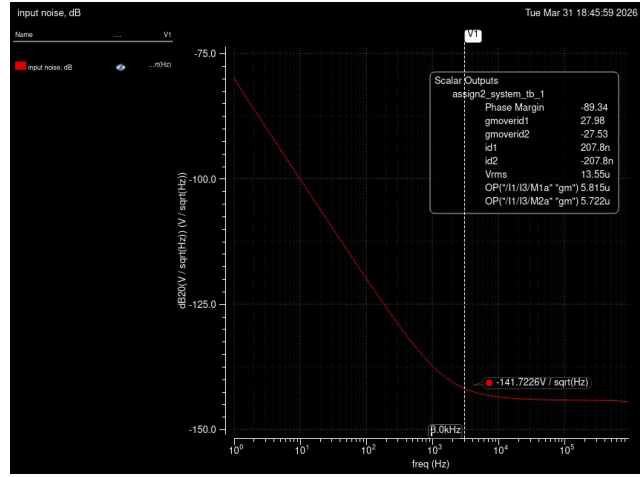


Figure 8: 1/f noise corner frequency

The simulated input-referred noise and output-referred noise are reported in *Figure 10*. It is worth noting the presence of distinct noise peaks at 20 kHz and 60 kHz. The dominant peak at 20 kHz corresponds to the chopping frequency, while the smaller peak at 60 kHz appears at three times the chopping frequency. This behavior is consistent with theoretical expectations; specifically, the inherent half-wave symmetry of the chopping signals effectively suppresses even-order harmonics, thereby explaining the absence of a spectral peak at 40 kHz.

Note that the input-referred noise is calculated via the expression:

$$V/\sqrt{\text{Hz}} = rfInputNoise("V/\sqrt{\text{Hz}}"?result"pnoise")$$

The Noise Efficiency Factor (NEF) of the chopper-stabilized amplifier was evaluated using the following expression:

$$NEF := V_{rms,in} \sqrt{\frac{2 \cdot I_{tot}}{\pi \cdot V_T \cdot 4kT \cdot f_{BW}}} = 1.71$$

The obtained input-referred noise corresponds to $V_{rms} = 4.769 \mu\text{V}$, which is consistent with state-of-the-art performance achieved through chopping techniques.

It is essential to highlight that the operational parameters from *Exercise 2* were maintained, specifically a bias current (I_{BN}) of approximately 450 nA. Under these consistent conditions, the current design achieved a better (smaller) Noise Efficiency Factor compared to the value of approximately 3 recorded in *Assignment 2*.

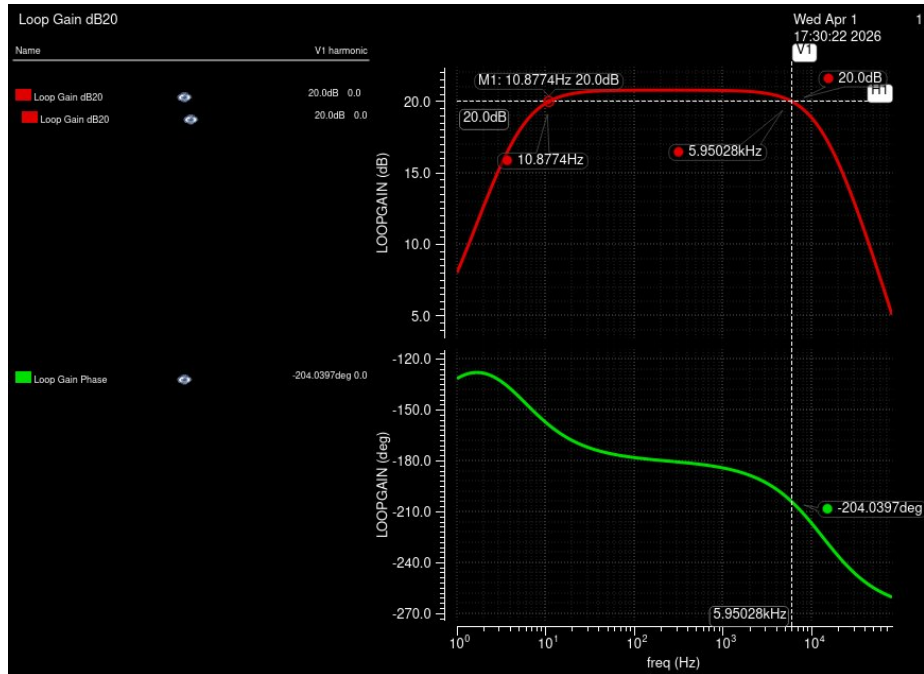


Figure 9: Bode plots: magnitude, phase

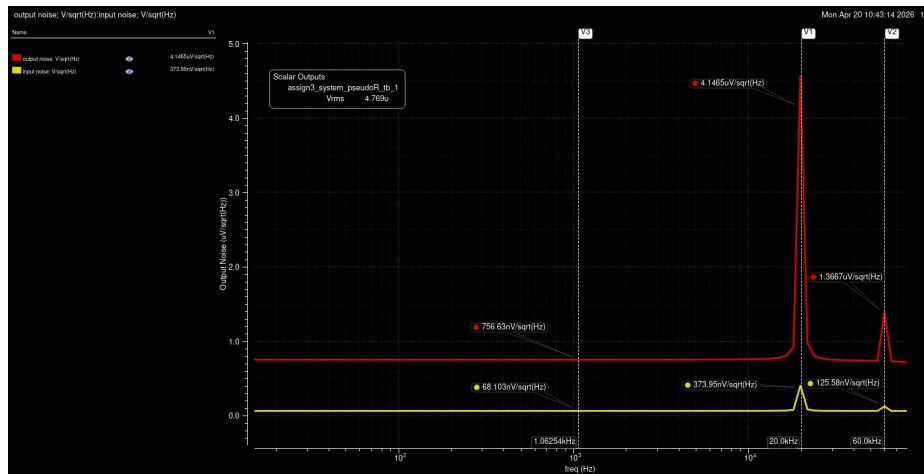


Figure 10: Input-referred noise and Output-referred noise

3 Task

3.a)

The following analysis evaluates how chopper stabilization mitigates the residual DC offset caused by a width mismatch in the NMOS input pair.

Unlike a standard amplifier where the input-referred offset ($V_{os,in}$) remains at DC, in a chopped architecture, the amplifier's internal offset is modulated solely by the output chopper. Consequently, this DC offset is shifted away from the baseband up to the chopping frequency f_{ch} and its odd harmonics ($3f_{ch}, 5f_{ch}, \dots$).

Without mismatch: The spectral peaks at f_{ch} are minimal, representing only the up-modulated $1/f$ noise.

With mismatch: These peaks increase significantly, as they now contain the amplified and up-modulated DC offset.

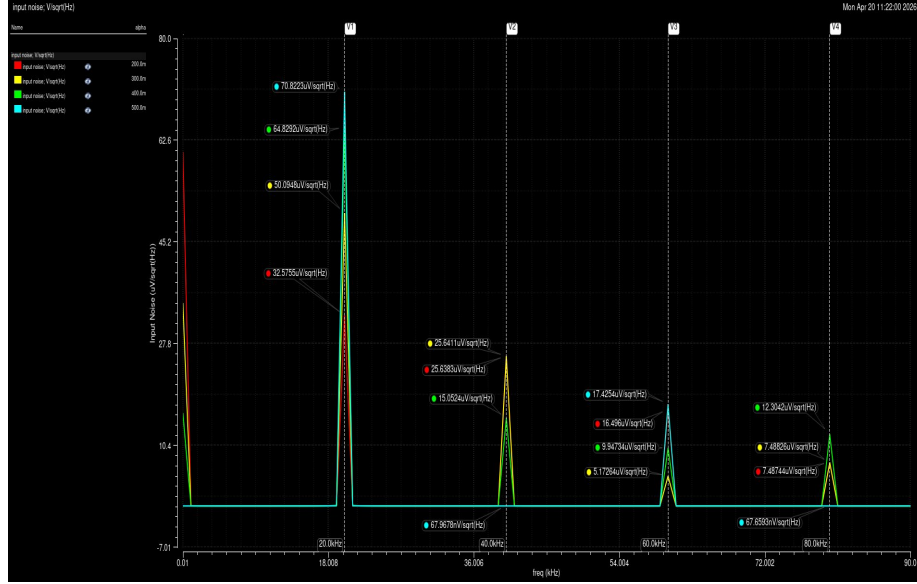
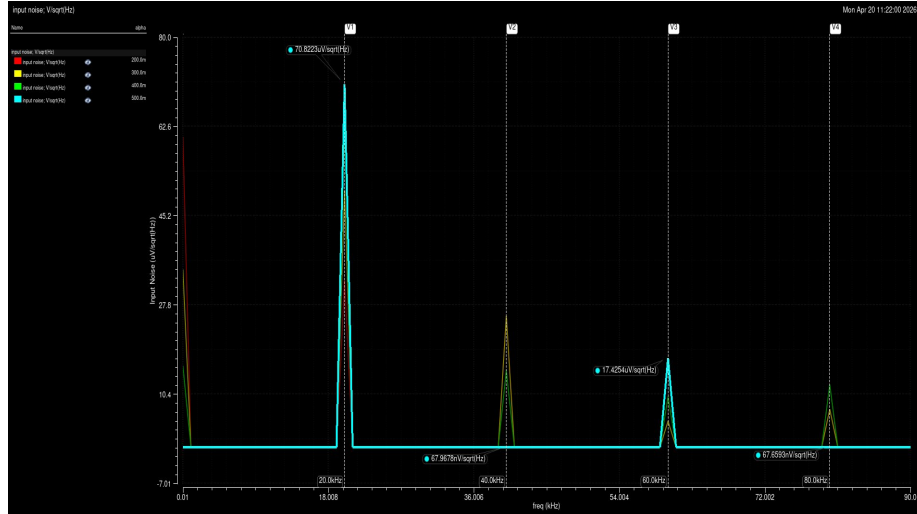
3.b)

The duty cycle α of the chopping signals is a critical parameter for the complete cancellation of the offset. By sweeping α from 50% down to 20%, the following observations are made regarding the residual offset. At a 50% duty cycle, the modulated offset signal is perfectly symmetrical, as the positive and negative halves of the square wave average to exactly zero, theoretically eliminating the DC offset.

As the duty cycle decreases ($\alpha < 0.5$), the symmetry of the modulation is broken and thus the offset signal spends more time in one polarity than the other; consequently, you will see the residual offset increase significantly as α moves further away from 0.5.

The simulation results confirm that a 50% duty cycle is mandatory for effective flicker noise suppression, as demonstrated in *Figure 11*. Any deviation of α from 0.5 induces noise leakage, causing a significant portion of the $1/f$ spectrum to remain at baseband.

Consequently, the spectral energy is no longer confined to the fundamental frequency and its odd harmonics ($f_{ch}, 3f_{ch}, 5f_{ch}, \dots$). Instead, the loss of half-wave symmetry introduces even-order harmonics ($2f_{ch}, 4f_{ch}, \dots$) into the frequency spectrum, as specifically highlighted in *Figure 12*.

Figure 11: Input-referred noise sweeping duty cycle $\alpha = 0.2 : 0.1 : 0.5$ Figure 12: Input-referred noise focus on $\alpha = 0.5$

4 Task

The input impedance (Z_{in}) of the chopper-stabilized amplifier is a critical parameter. To counteract the reduction in input impedance caused by the switching action of the chopper, a capacitive positive feedback loop is implemented as observed in *Figure 13* and *Figure 14*. This technique effectively cancels the input capacitance by feeding back a portion of the output signal to the input node; ensuring that the amplifier's input impedance remains significantly larger than the sensor's source impedance, thereby preventing signal attenuation due to loading effects.

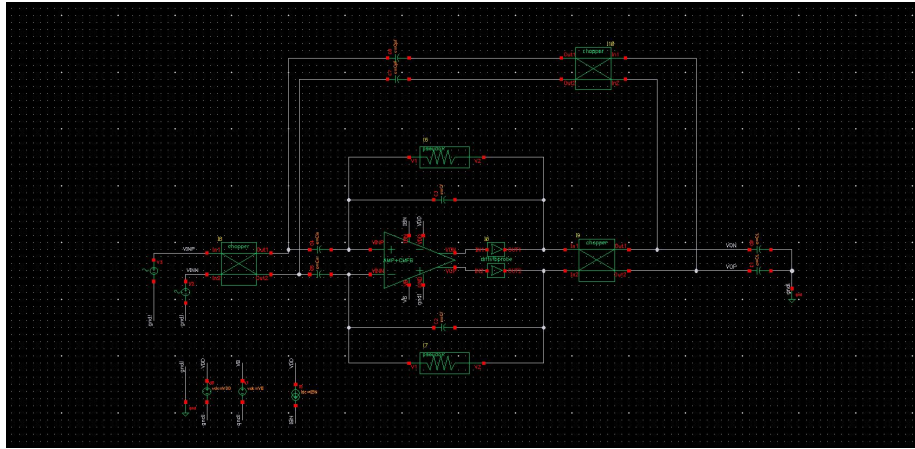


Figure 13: System with pseudo-resistors, choppers and input-boosting loop

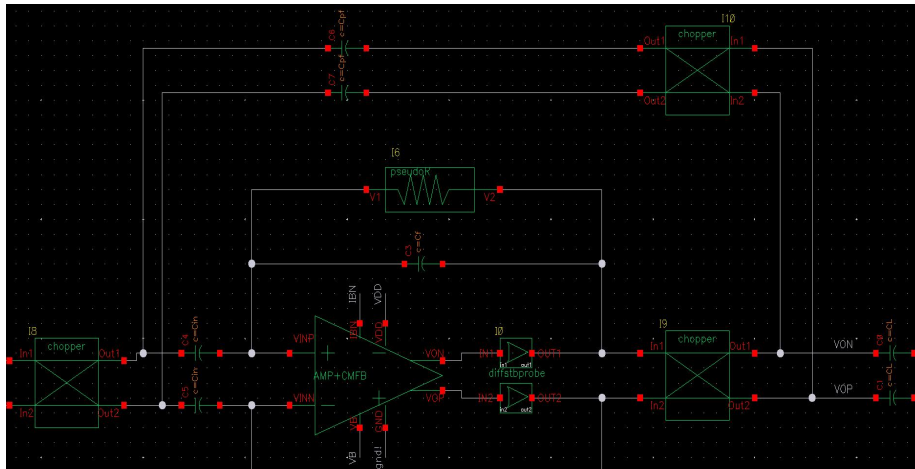


Figure 14: Zoom on input-boosting loop

The required positive feedback capacitance C_{pf} , to achieve optimal cancellation, is determined by the relationship:

$$C_{pf} = \frac{C_{in} \cdot C_f}{C_{in} - C_f} = 435 \text{ fF}$$

The differential input impedance with positive feedback is first calculated in the *Cadence Virtuoso calculator* using the following expression, which accounts for the differential voltages and currents from the Periodic AC (PAC) analysis:

```
Zin_with_pf = ((vh('pac "/VINP" (harmonic = 0)) - vh('pac "/VINN"
(harmonic = 0))) / (ih('pac "/V3/PLUS" (harmonic = 0)) - ih('pac
"/V2/PLUS" (harmonic = 0))))
```

To simplify the extraction of the differential input impedance Z_{in} , a differential test voltage source 1 V is applied to the input nodes. In this specific configuration, the input impedance is derived directly from the reciprocal of the total differential input current at the fundamental frequency:

$$Z_{in} = \frac{\Delta V_{test}}{\Delta I_{in}} = \frac{1V}{I_{in}(\text{harmonic } 0)}$$

As illustrated by the comparison between *Figure 15* (un-boostered input impedance) and *Figure 16* (with positive feedback loop), a significant enhancement in the input impedance Z_{in} is observed. The implementation of the capacitive positive feedback loop yields an increase of approximately **190x** relative to the baseline configuration, demonstrating a highly effective bootstrapping mechanism. While this enhancement is substantial, it is heavily frequency-dependent. The primary reason for the impedance drop at higher frequencies is the phase delay introduced by the amplifier. Because the positive feedback is derived from the amplifier's output, the finite bandwidth of the OTA causes the output signal to accumulate a phase shift as the signal frequency increases. Consequently, the charge injected by the feedback network is no longer perfectly in-phase with the input signal, preventing an exact cancellation of the input current. As a result, the 190x impedance boost is only maintained across a narrow bandwidth, primarily up to approximately 200 Hz. Nevertheless, as highlighted by the marker in *Figure 16*, the input impedance at 100 kHz remains at approximately 2.23 MΩ. It is noteworthy that even at this higher frequency, the boosted impedance value is comparable to the low-frequency impedance of the original amplifier without the input boosting loop, ensuring that the loading effects remain minimized even outside the primary signal bandwidth.

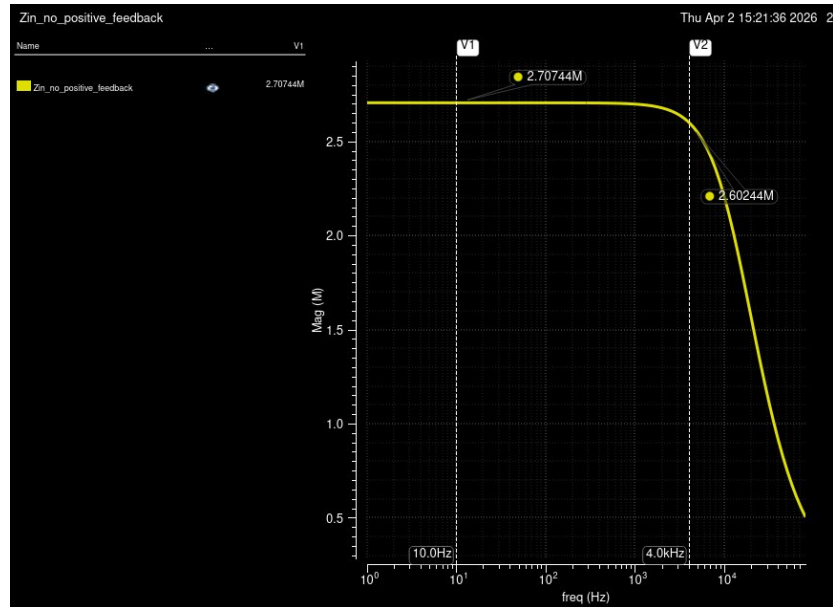


Figure 15: Input impedance without positive feedback

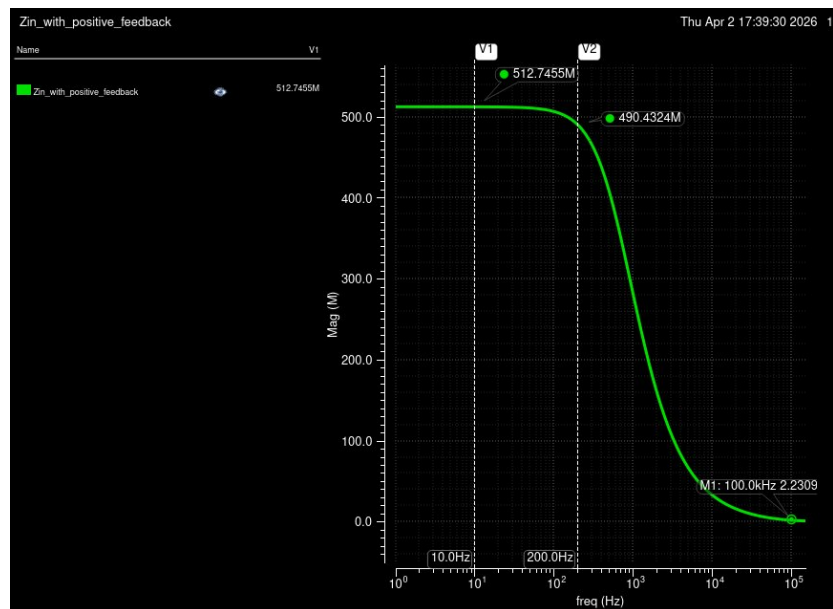


Figure 16: Input impedance with positive feedback

Energy-efficient Analog Integrated Circuits

Sensor Interface – Exercise 3

Assignment

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Prof. Taekwang Jang

1. Consider the inverter-based amplifier with capacitive feedback from Exercise 2 - task 3.
 - a. Replace the feedback resistance R_F with the pseudo-resistor configurations seen in the lecture.
 - b. Discuss advantages and disadvantages of each implementation. Support your discussion using simulation results.

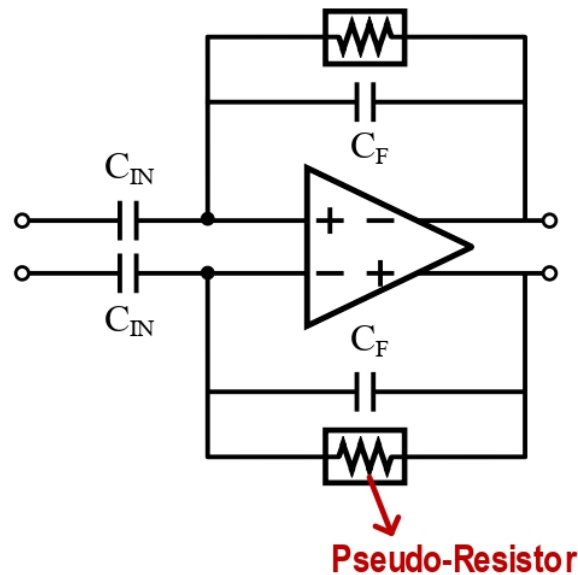


Fig. 1: Amplifier with capacitive feedback

2. Introduce chopping to eliminate low-frequency artifacts. Run periodic AC (pac), periodic noise (pnoise) and stability (pstab) analyses to simulate the transfer function, input-referred noise and noise efficiency factor of your chopper-stabilized amplifier. Assume a load capacitance of 100 fF and choose the chopping frequency considering that a signal bandwidth f_{BW} from 100 Hz to 5 kHz is required.

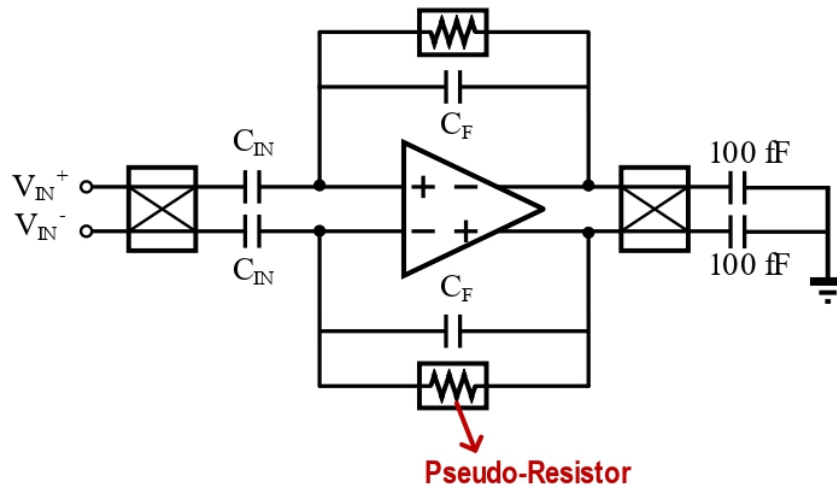


Fig 2: Chopper-stabilized amplifier

3. Consider the amplifier from Exercise 2 (question 4), where you introduced a mismatch in the width of the n-channel input pair.
 - a. Check the effect of chopping on the amplifier input-referred offset.
 - b. Sweep the duty cycle α of your chopping signals from 20% to 50% (Fig. 3). How does the duty cycle affect the residual offset?
4. Simulate the input-impedance of your chopper-stabilized amplifier using transient or pac analysis. Implement a capacitive positive feedback loop to boost the input impedance (Fig. 4).

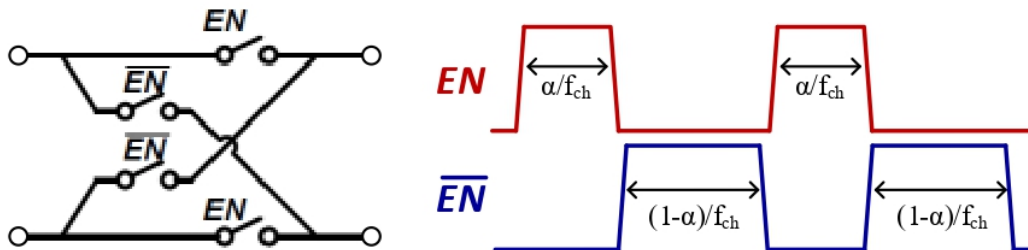


Fig 3. Chopping switches and waveforms

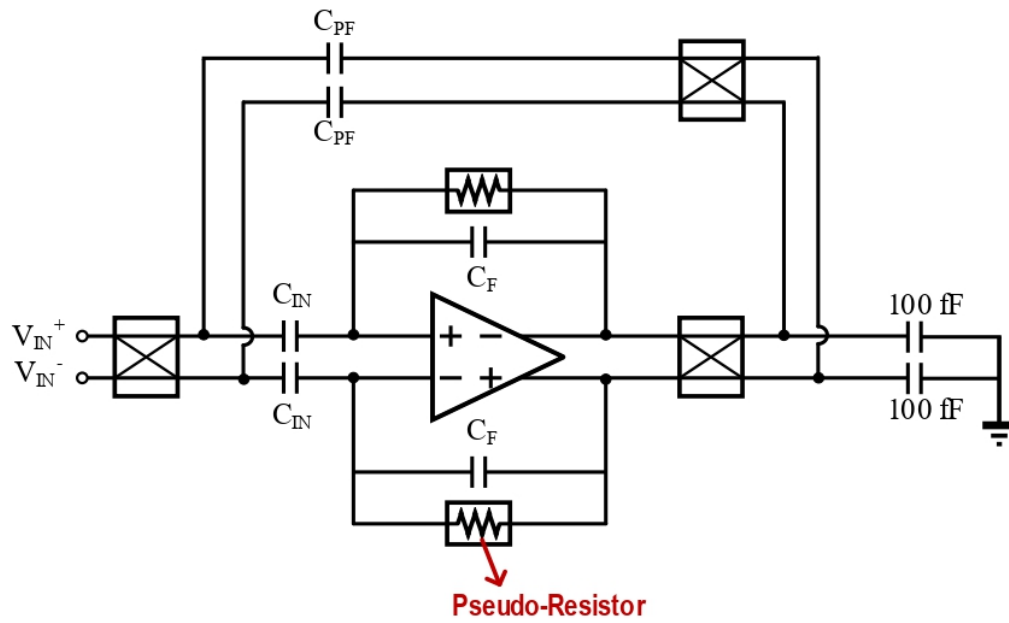


Fig 4: Chopper-stabilized amplifier with impedance-boosting loop